

Individual Addendum
QM 2023 Capstone Project - Milestone 4

Name: Nicholas Langkamp

Team: 2nd Row Team

Date: 1 May 2026

PERSONAL CONTRIBUTIONS

M1 (~3 Hours)

- I helped other teammates figure out how to get a pipeline code written
- I gave ideas for how we should set up the research and hypothesize.

M2 (~5 Hours)

- I helped generate most charts, focusing more on the visualization than the interpretation
- I created the HTML dashboard and edited it to get formatting better

M3 (~10 Hours)

- I helped get the code ready for regressions
- I met with Dr. Seagraves twice to get direction on the milestone to rework part of it
- I redid the regressions for the firm-specific data in M3v2, created all M3v2 results and figures, and created an HTML dashboard

M4 (~5 Hours)

- I helped write the memo as a whole, insert figures, and generate the powerpoint
- I hand-wrote the executive summary in the memo and edited the slides

Total hours across all milestones: 23 hours

Percentage of team workload: 40%

Roles on team: In charge of regressions and data presentation. Took the lead on M3.

DEFENDED DECISION

While making M3, I realized that the regressions were not giving significant results. After consulting with the professor, I advocated for adding firm-level data to the panel to see more significant trends and be able to subdivide the data into more segments. It allowed us to see the impact on smaller firms.

KEY LIMITATIONS

Sentiment has limited predictive power on market returns, but a key limitation is that the market returns have a high impact on sentiment. When the market is down, consumers have worse sentiment towards the economy. This makes the data almost circular, and outside context is necessary.

AI AUDIT NOTES

I personally updated the team AI Audit Appendix for all AI outputs I prompted that were included in the final repository. I used Github Copilot to gather context outside of deliverables to help me understand confusing issues, but no results of such queries were used in the capstone project.